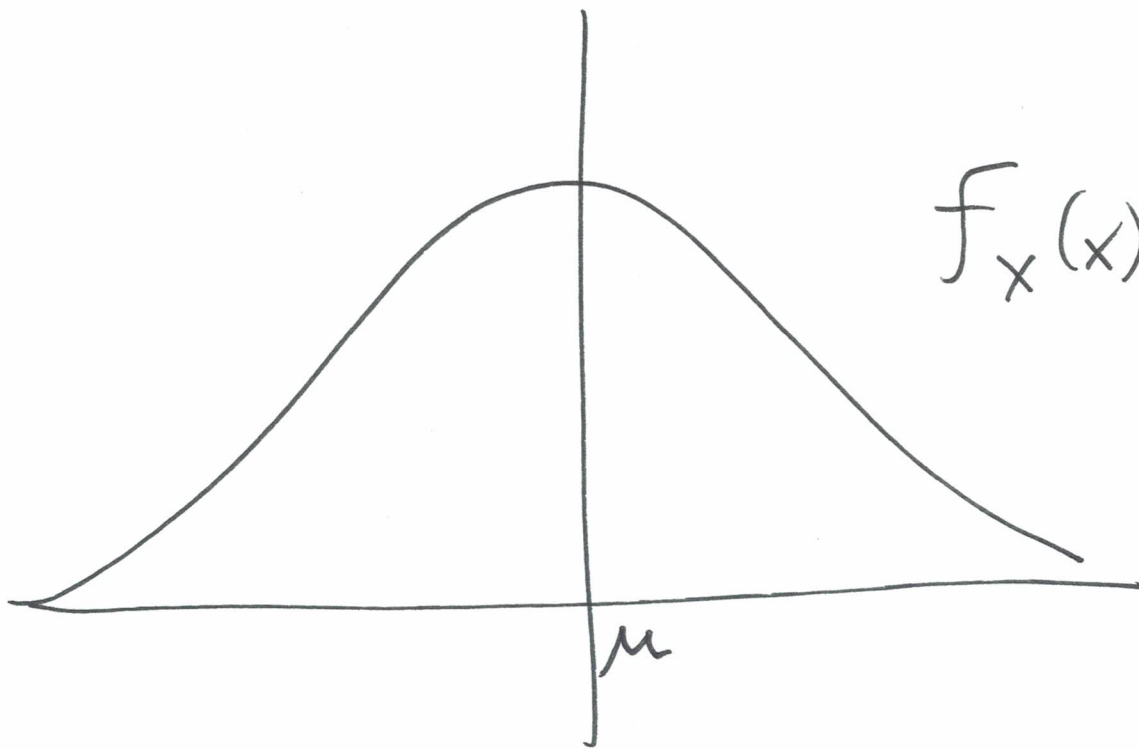


THE GAUSSIAN DISTRIBUTION

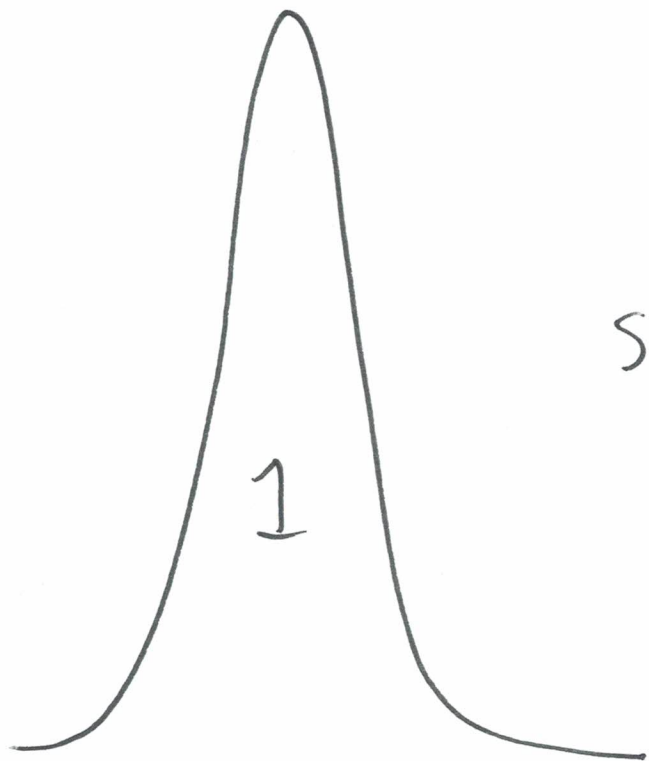


$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

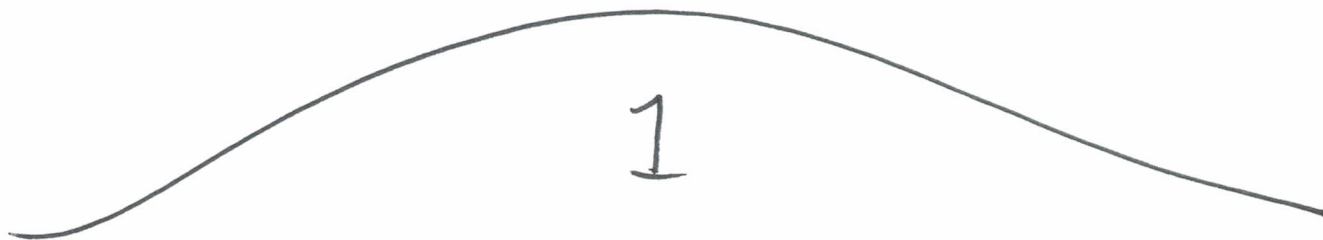
μ = MEAN

σ = STANDARD
DEVIATION

σ^2 = VARIANCE



SMALL σ



LARGE σ

WHY IS THIS A VALID PDF?

$$\int_{-\infty}^{\infty} f_X(x) dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$x \leftarrow x - \mu$$

$$\downarrow \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}} dx$$

$$x' = \frac{x}{\sigma} \quad dx' = \frac{1}{\sigma} dx$$

$$\downarrow \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x'^2} dx'$$

$$K = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx$$

TRICK

$$K^2 = \left(\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx \right) \left(\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}y^2} dy \right)$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x^2+y^2)} dx dy$$

$$r^2 = x^2 + y^2$$

$$r dr d\theta = dx dy$$

$$\downarrow \begin{matrix} 2\pi & \infty \\ 0 & 0 \end{matrix} \quad \frac{1}{2\pi} \int \int e^{-\frac{1}{2}r^2} r \, dr d\theta$$

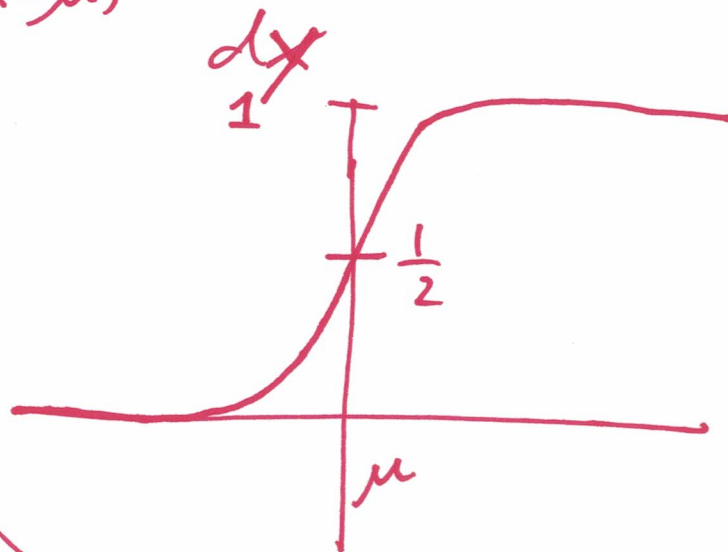
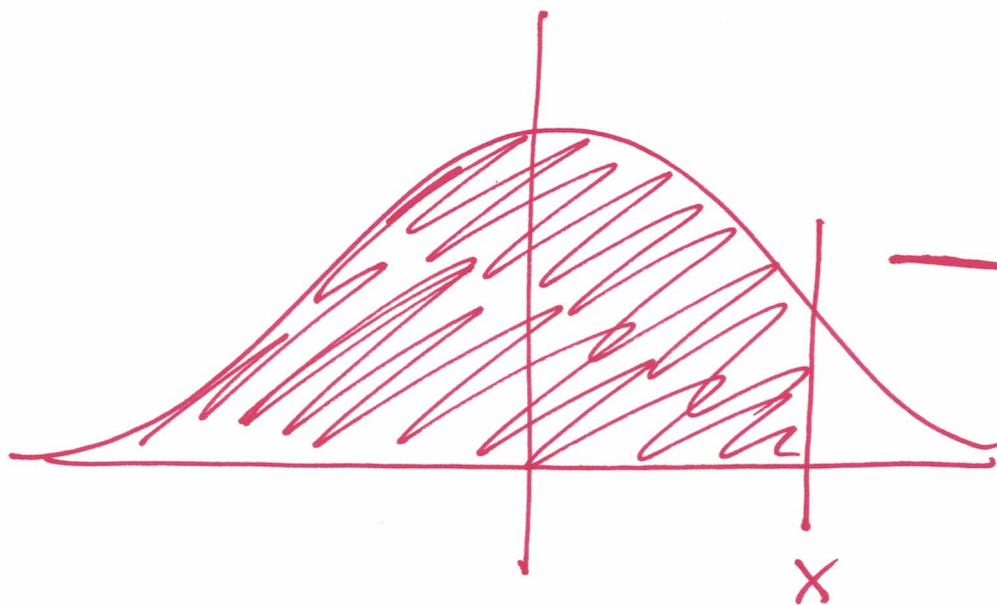
$$\rightarrow \int_0^{\infty} r e^{-\frac{1}{2}r^2} dr = -e^{-\frac{1}{2}r^2} \Bigg|_{r=0}^{r=\infty}$$

$$= 0 - (-1) = 1.$$

CDF OF GAUSSIAN:

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(y) dy$$

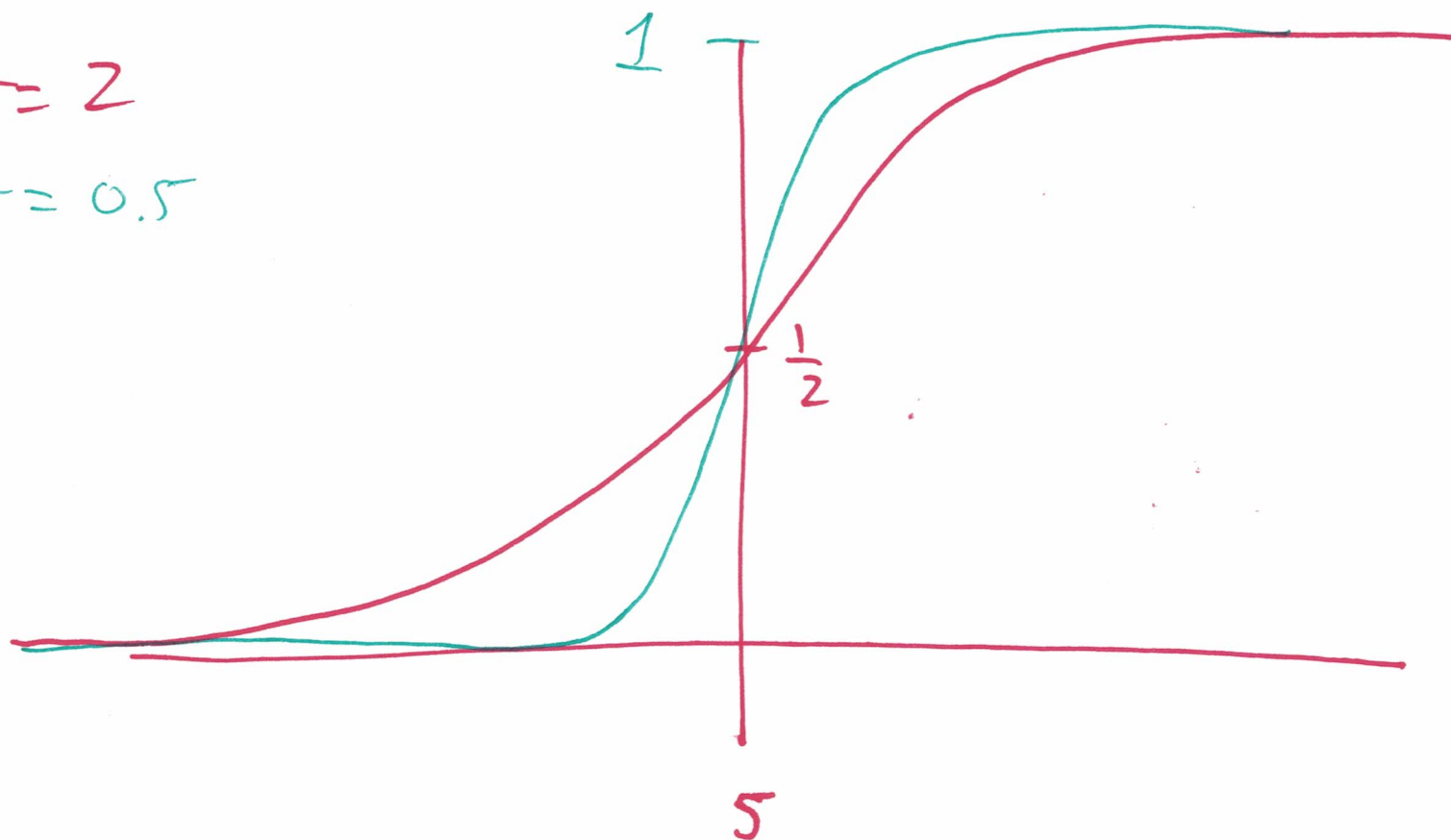
$$= \int_{-\infty}^x \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(y-\mu)^2} dy$$



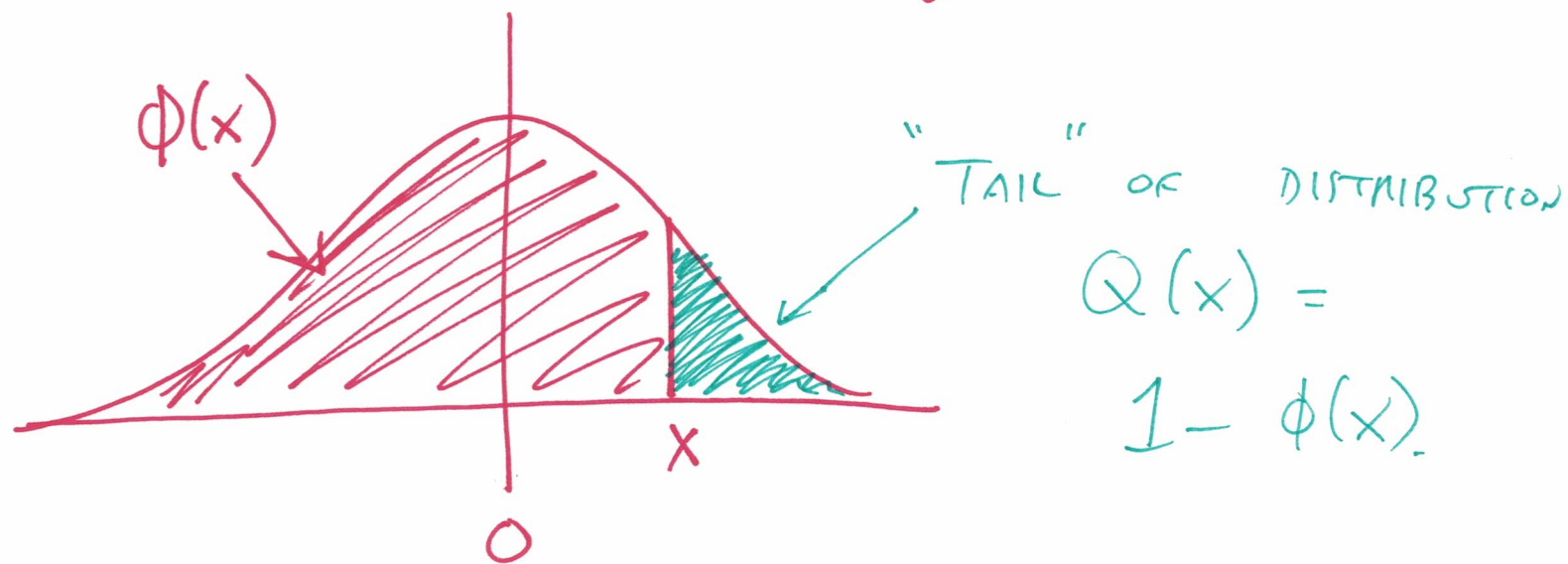
$$\mu = 5$$

$$\sigma = 2$$

$$\sigma = 0.5$$



ANY GAUSSIAN CDF PROBLEM CAN BE
TURNED INTO A QUESTION ABOUT
THE "STANDARD NORMAL" $\mu=0 \quad \sigma=1$



$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{y^2}{2}} dy$$

How TO TRANSFORM TO THE
STANDARD NORMAL:

$$P(X \leq x) = P\left(\frac{x - \mu}{\sigma} \leq \frac{x - \mu}{\sigma}\right)$$

↑
ANY
GAUSSIAN
 μ, σ

$$= P\left(Z \leq \frac{x - \mu}{\sigma}\right)$$

↑
GAUSSIAN
WITH $\mu = 0$
 $\sigma = 1$

"HOW MANY
STANDARD
DEVIATIONS
FROM
THE
MEAN?"

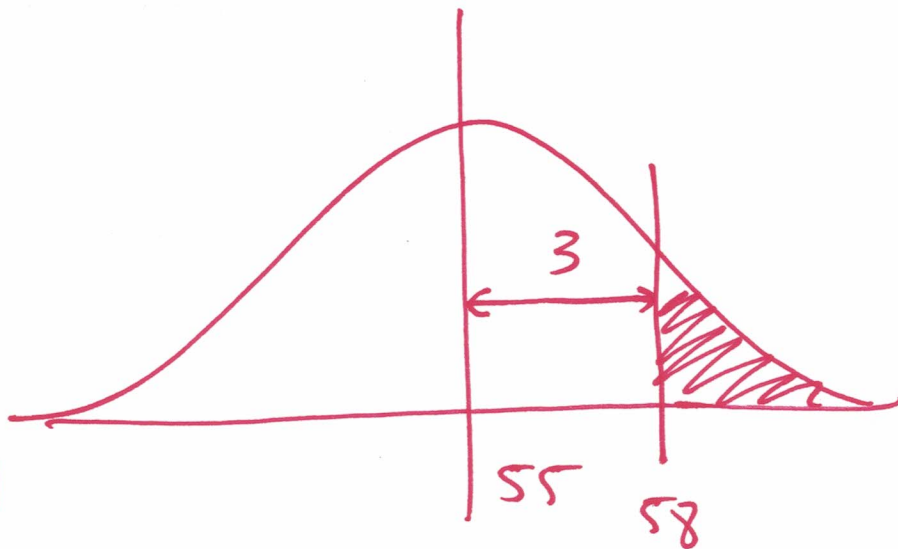
⇒ Q TABLE

X GAUSSIAN $\mu=55$, $\sigma=3$

$$P(X > 58)$$



$$P\left(\frac{X-55}{3} > \frac{58-55}{3}\right)$$



$$P(Z > 1) = Q(1) = 0.159.$$



STANDARD
NORMAL

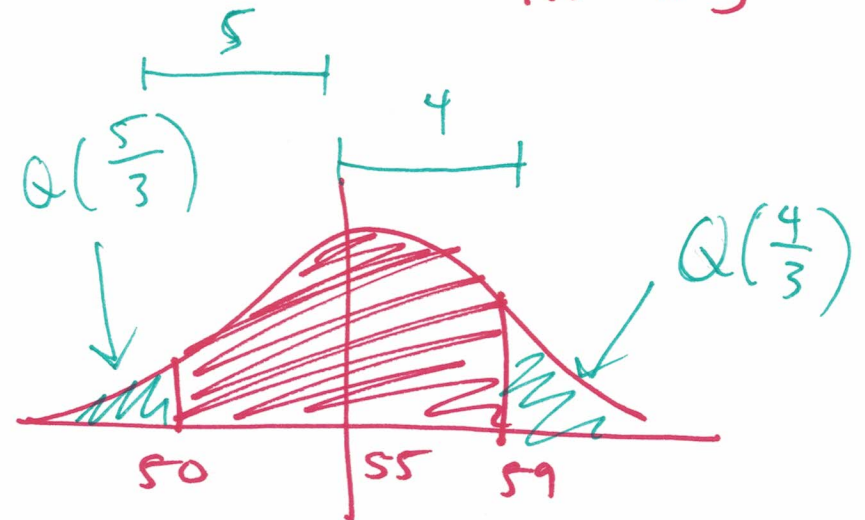
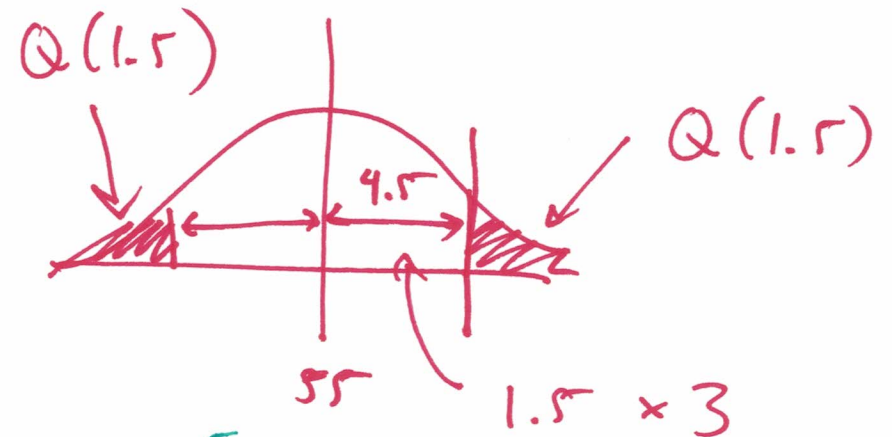
$$P(|X - 55| > 4.5)$$

"

$$2Q(1.5) =$$

$$2(0.0668) = \dots$$

$$P(X \in [50, 59])$$



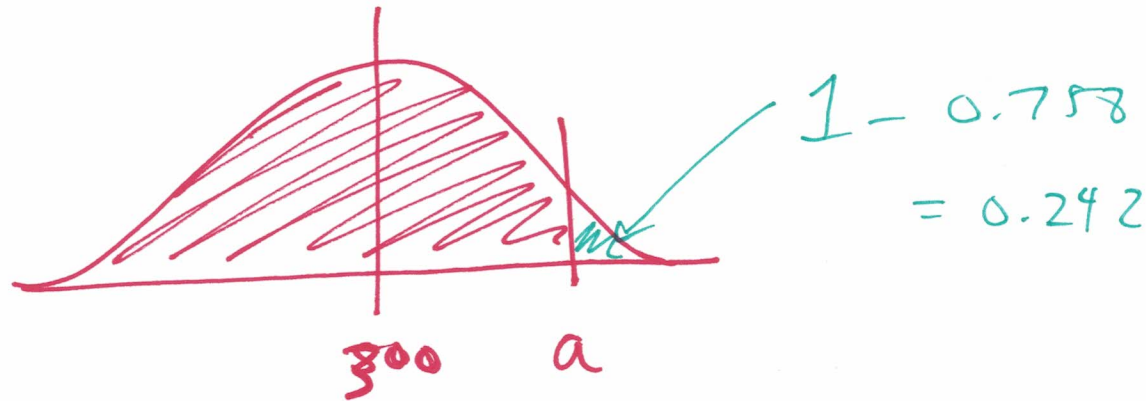
$$1 - Q\left(\frac{5}{3}\right) - Q\left(\frac{4}{3}\right)$$

$$= 1 - Q(1.66\dots) - Q(1.33\dots)$$

$$\approx 1 - \overset{0.0478}{Q(1.7)} - \overset{0.0912}{Q(1.3)} \dots$$

$W \quad \mu = 300 \quad \sigma = 50.$

FIND a SO THAT $P(W < a) = 0.758.$

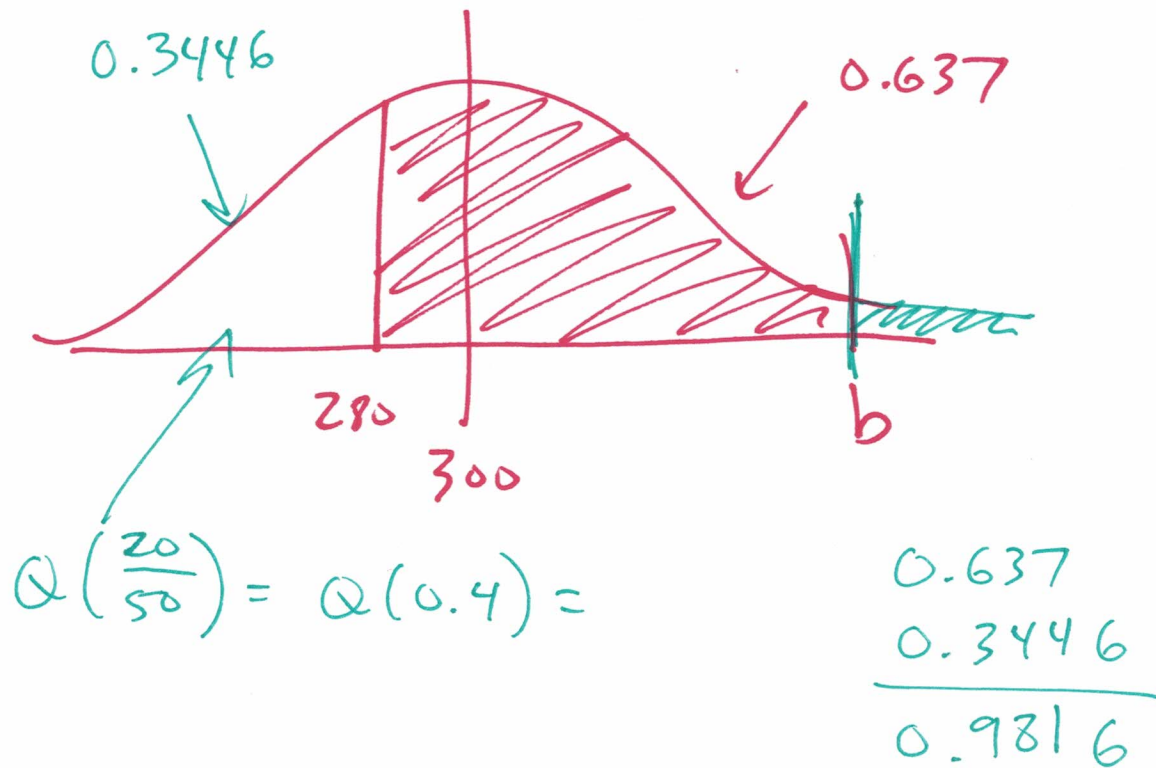


Z FOR STANDARD NORMAL SO $Q(z) = 0.242.$

$Q(0.7) = 0.242.$
FROM TABLE.

$a = 300 + 0.7(50) = 335.$

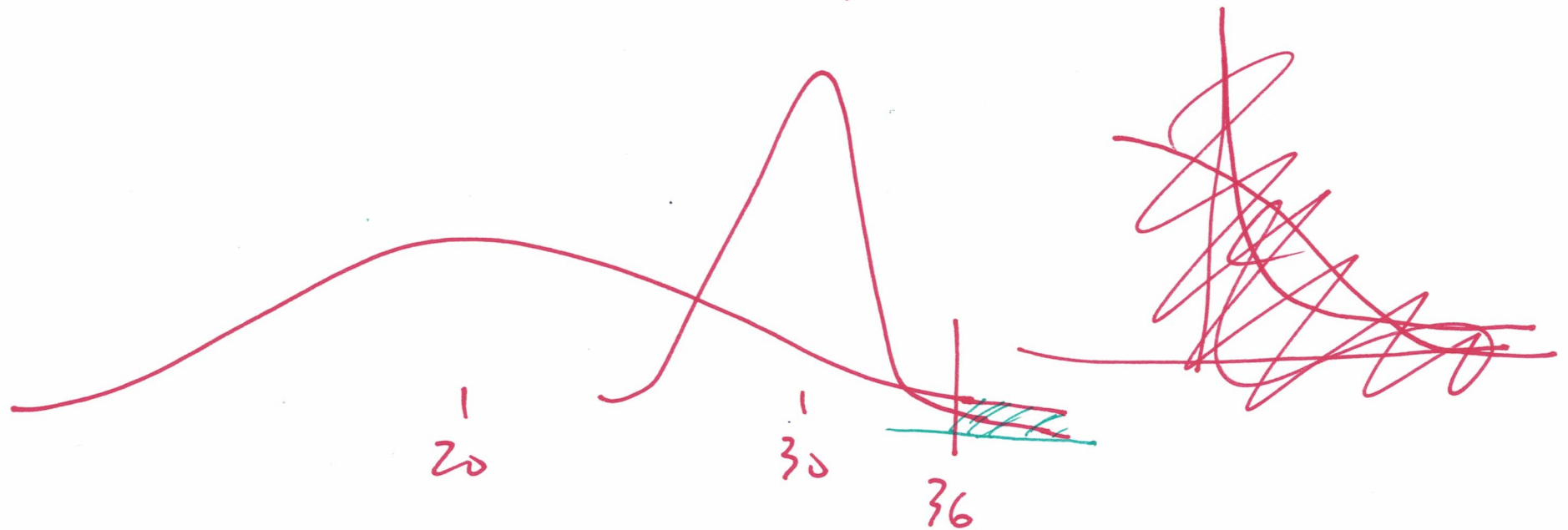
$$P(280 < w < b) = 0.637$$



$$Q(Z) = 0.0184 = 1 - 0.9816.$$

$\approx Z \approx 2.1$ FROM TABLE.

TF2	SCOUT	LIFETIME	$\mu = 30$	$\sigma = 2$
	SPY	LIFETIME	$\mu = 20$	$\sigma = 6$



WHO IS MORE LIKELY TO SURVIVE AT LEAST
36 SEC?

SCOUT: $\frac{6}{2} = 3$ σ 'S AWAY.

SPY: $\frac{16}{6} = \frac{16}{6}$ σ 'S AWAY. 2.667. $\leftarrow$ MORE AREA UNDER TAIL.

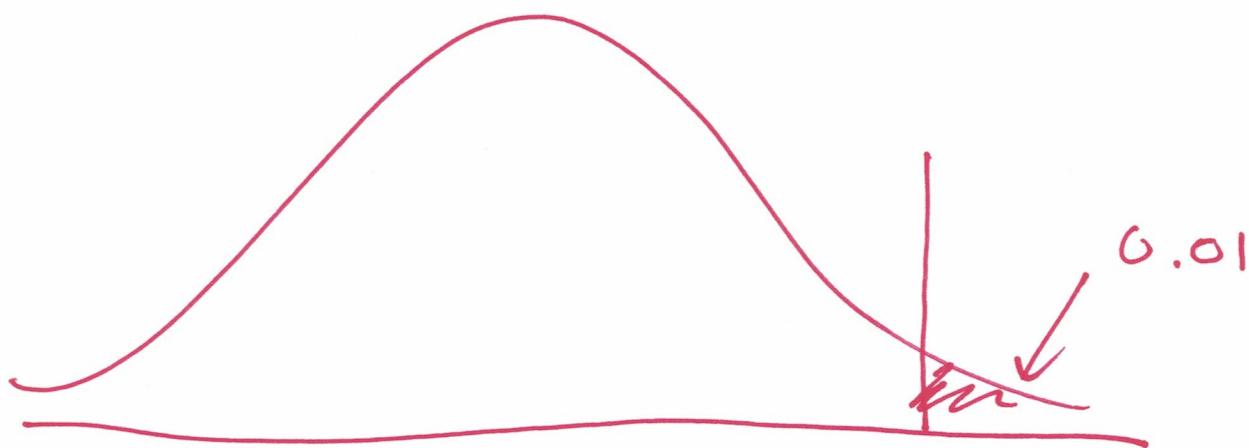
WHAT IS t SO THAT THEY ARE
EQUALLY LIKELY TO SURVIVE AT LEAST t SEC?

$$t = 30 + 2\sigma = 20 + 6\sigma$$

$$10 = 4\sigma$$

$$\sigma = 2.5.$$

$$35 \text{ SEC.}$$



$$Q(z) = 0.01$$

z	$Q(z)$
2.32	0.01
	0.001

"INVERSE
Q
TABLE"

$$Q(z) = 1.7$$

$$0.17$$

$$1.4$$

$$0.0184$$

$$0.0157$$

$$2.3$$

$$z = 0.95$$

EXAM 2:

THURSDAY MARCH 26

IN CLASS

FINAL EXAM:

THURSDAY MAY 7 3-6 PM.

NO OFFICE HOURS NEXT WEEK

(SPRING BREAK!)